

LEO

Local Energy Orchestrator — neighbourhood-scale flexibility that keeps clean power dependable on India's most stressed distribution transformers

Yuva Yodha Energy Tech Hackathon · Challenge: **Grid Reliability — Renewable Intermittency**

Submission document: solution write-up, architecture, design artefacts, simulation, measured reliability improvement, ownership & O&M model and unit economics

Target deployment: an overloaded 100 kVA distribution transformer in peri-urban Karnataka (BESCOM area)

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1. Executive summary

India's distribution transformers (DTs) are where renewable intermittency and growing evening demand collide. Rooftop solar exports at midday and disappears at sunset exactly as cooling and lighting load peaks; on the Indian Energy Exchange this “duck curve” takes evening prices to the ₹10/kWh ceiling while midday blocks clear near ₹1.91/kWh^[1, 2, 3]. Overloaded DTs fail: BESCOM lost 38,288 of 497,991 transformers (7.96%) in FY 2023-24, 29% of them to overload^[4, 5]. Regulators have now made flexibility an obligation — Karnataka's DF/DSM Regulations 2026 require DISCOMs to procure demand flexibility of 0.5% of peak, rising to 2.0% by FY 2029-30, on the Maharashtra model of a ₹2,000/kW-yr shortfall penalty^[6, 7, 8].

LEO is a software-and-small-hardware coordination layer that a local cooperative or NGO, registered as a demand flexibility aggregator, runs at the DT. It forecasts tomorrow's load and solar per phase, plans three single-phase second-life battery blocks and two automated demand-response programmes around the forecast, corrects live from its own low-cost sensors, protects critical premises during outages through a small backup circuit, and settles everything transparently a day later when smart-meter data arrives. It commands only assets the operator owns; everything on the DISCOM network is a recommendation.

We built the whole system as a running prototype and evaluated it on a simulated year — one week in every month — of a real Hoskote (Bengaluru Rural) street grid with 150 households on a 100 kVA DT, against an identical baseline without LEO.

129 → 72 kW average evening peak at the DT (44% lower; rating ≈ 95 kW)	0% → 100% critical premises (clinic, water pump, school, shops) powered through outages and load shedding	5.8% → 0.1% expected DT failure rate per year (IEEE C57.91 ageing); 414 customer-hours/yr of failure outages avoided
60.9 kW verified peak reduction for the DISCOM's DFPO obligation	₹5.03 L/yr DISCOM gross saving per DT (₹50.32 Cr/yr per 1,000 overloaded DTs)	5.0 years operator payback; 10-year NPV ₹1.10 L; households pay ₹0 upfront

The economics close because three value streams stack on an overloaded DT: evening power the DISCOM no longer buys (₹3.52 L/yr), DFPO penalty avoided (₹1.22 L/yr) and transformer failures avoided (₹29,003/yr). A two-part contract — the ₹2,000/kW-yr DFPO benchmark plus ₹5.00 per verified evening kWh — falls inside a deal zone (₹4.56–₹5.77/kWh) where the operator, the DISCOM and households all come out ahead. Every price in the model is sourced from public reports, tariff orders and Indian pilots, or flagged for verification.

2. Problem statement

The challenge asks for affordable, neighbourhood-scale flexibility tools — forecasting, smart load management, demand response, shared storage and local coordination — that keep electricity dependable when renewable supply falls below demand, stay local and manageable, are viable for low-income urban and peri-urban communities, and support the DISCOM^[9]. Four facts about Indian distribution grids shape our answer.

2.1 The evening ramp and the duck curve

Solar generation drives day-ahead prices to their floor at midday — average lowest blocks around ₹1.91/kWh and sometimes below ₹1 — and collapses at sunset, when residential cooling and lighting surge. Evening prices routinely hit the ₹10,000/MWh regulatory ceiling, an average evening premium of about ₹7.92/kWh and a midday-to-evening spread above ₹8,000/MWh^[1, 2, 3, 10]. The intermittency gap is therefore daily and predictable, not only a weather event.

2.2 Transformers fail from thermal overload

BESCOM's DT failure rate has stayed between 7.5% and 8.07%; in FY 2023-24 it lost 38,288 of 497,991 units (7.96%), and sustained overload causes 29.09% of failures^[4, 5]. Overloading raises hot-spot temperature and accelerates insulation ageing, cutting an expected 30-year life to 11-15 years; one Indian case study recorded failures within 20-28 months^[11, 12]. A 100 kVA unit costs about ₹5.05 lakh to replace; a repair about ₹11,520^[13, 14]. Peri-urban DTs that grew with their neighbourhoods without augmentation are the ones at risk — and a failed DT means a multi-day outage for every household on it.

2.3 Losses peak with the peak

BESCOM's AT&C losses fell to 9.13% in FY 2023-24^[15]. The technical part is resistive (I^2R) and grows with the square of current, so a disproportionate share is lost precisely at the evening peak; single-phase heavy appliances make Indian LV networks unbalanced, adding neutral current^[11, 14].

2.4 The DISCOM is blind below the feeder, and household data arrives a day late

DISCOMs see the 11 kV feeder through SCADA and the National Feeder Monitoring System^[16] but only about 3% of DT meters were communicating in 2024^[17]. Smart meters are rolling out under RDSS (over 5.28 crore installed)^[18], but the contractual AMISP service level is that 95% of 15-minute data reaches the DISCOM within 8 hours and 98% within 12 hours^[19, 20, 21]. Any design that assumes live household data in India today will not work in the field.

2.5 Flexibility is now an obligation

Maharashtra's 2024 DF/DSM Regulations created Demand Flexibility Portfolio Obligations (DFPO): 2.5% of the previous year's peak, rising to 7.0% by FY 2029-30, measured in kW at a single instance in the peak period, with a ₹0.20 crore/MW (₹2,000/kW) disincentive for shortfall and an equal incentive for over-achievement, and they formally recognise aggregators^[7, 8, 22, 23]. Karnataka followed in March 2026 with targets of 0.5% rising to 2.0% and verification by empanelled independent agencies^[6, 24, 25]. DISCOMs therefore need a

source of verifiable, local kW — and a low-income neighbourhood with an overloaded DT is where that kW is worth the most.

3. Alignment to the selected challenge

We chose **Grid Reliability — Renewable Intermittency**. LEO touches four of the illustrative solution areas named in the brief: shared storage with fair-access pooling (second-life batteries), local control with smart load management, demand response and critical-load priority, pooled reliability across users, and a DISCOM-facing dashboard that predicts feeder stress and recommends actions^[9].

Table 1. Traceability from the brief to LEO.

Brief requirement	How LEO answers it	Evidence in this document
Bridge intermittency gaps	Midday solar stored in three per-phase blocks and released over the evening peak; pumps moved into the solar window daily; AC events on stress days; Pre-outage for announced load shedding	Evening peak 129→72 kW; 49,836 kWh/yr of local solar stored; critical premises 100% powered
Stay local and manageable	One DT, one edge gateway that keeps running offline, LoRa sensors with no SIM cost, one technician per 20 DTs; no DISCOM equipment is commanded	Operator controls only its own assets; DISCOM receives recommendations
Genuinely affordable	Second-life LFP packs; ₹999 IR blasters and ₹2,090 smart plugs instead of new appliances; households pay nothing upfront and bills are unchanged	Operator payback 5.0 yr; pump homes earn ≈₹2,027/yr, AC homes ≈₹1,774/yr
Support the DISCOM	Verified kW for DFPO; day-ahead stress forecast; structured recommendations (tap change, outage scope); fewer transformer failures; lower peak purchase and losses	60.9 kW verified; DISCOM gross saving ₹5.03 L/yr per DT; losses –1,529 kWh/yr
Measurable reliability improvement vs a defined baseline	Identical simulated world with and without LEO, 84 days spanning all seasons, metrics M1–M5 plus outage hours and transformer ageing	Section 11
Ownership, O&M and unit economics	Cooperative/NGO aggregator running a cluster of 20 DTs; two-part DFPO contract; household revenue share	Sections 12–13

4. Proposed solution

LEO is the missing coordination layer between the DISCOM's feeder-level systems and the households on one transformer. It complements rather than competes with utility ADMS/DERMS: we adopt the DERMS idea of aggregating many small resources into one controllable virtual resource, but without claiming utility-grade control authority^[26, 27].

4.1 What is installed at one transformer

- **Six voltage/outage sensors** on LoRaWAN — three at the DT's LV busbar (one per phase, each with a split-core CT) and three at the far end of each phase^[28, 29, 30].
- **Three single-phase storage blocks**, one per phase: 60 kWh second-life LFP pack with BMS and a 15 kW hybrid inverter with a grid port and a backup port. Three blocks rather than one three-phase unit because LV problems are per phase^[28].
- **A backup circuit** — a separate LV cable from the battery backup ports to six registered critical premises (a clinic, a borewell pump, a tuition centre and three shops), each with a dual-source meter that enforces a current limit.
- **Automated demand-response devices** in enrolled homes: a 16 A smart plug on the water pump, a Wi-Fi IR blaster on the AC^[31].
- **An edge gateway** (Raspberry Pi-class, on UPS) that keeps planning, live control and outage response running if the internet link drops; and an **operator cloud** for forecasting training, DR, settlement, recommendations and the consoles.

4.2 What LEO does every day

1. **Forecast (D-1, 19:30)**. Per-phase P10/P50/P90 net load for the next 24 h from weather, calendar and day-late meter history.
2. **Model the network**. Run a three-phase power flow on the forecast: which bus, which hour, how far outside limits, and how much each battery may safely charge or discharge.
3. **Plan storage and DR**. Fill each block in the midday trough (solar export first) and spread its energy over the evening peak; schedule enrolled pumps into the solar window; on the year's most stressed days, call an automated AC event and send SMS offers for the residual.
4. **Operator approves (D-1, 23:30)** in the console, where every action carries its reason.
5. **Live correction**. Every 15 minutes, re-solve the battery setpoint from the busbar CT and sensors; absorb overvoltage at midday; on a load-shedding notice enter Pre-outage; on grid loss run the backup circuit.
6. **Escalate**. Residual violations become structured recommendations to the DISCOM (for example, a tap change).
7. **Settle (D+1)**. When meter data arrives, verify reductions against IPMVP Option C baselines and a random holdout, pay DR participants first, then pool the household share of revenue^[32, 33].

4.3 Who runs it and who pays

A local cooperative, NGO or self-help-group federation registers as a demand-flexibility aggregator — the role the MERC and KERC regulations recognise^[6, 7, 23] — and runs LEO on a cluster of about 20 DTs with one technician. The DISCOM pays it under a two-part contract: the DFPO capacity value for verified kW plus a payment per verified evening kWh. Households

pay nothing upfront, their DISCOM bills are untouched, and they are paid from the operator's realised revenue; there is no penalty anywhere in the system. Section 12 details the model.

5. Key features, user journeys and applications

5.1 Key features

Table 2. LEO's main features.

Feature	What it does	Why it matters
Day-ahead stress forecast	Per-phase P10/P50/P90 load and solar, run through a three-phase network model	Overloads and voltage problems are predicted a day early, with the reason (load, temperature, irradiance)
Peak-shaving storage	Water-filling plan, re-solved every 15 minutes from the busbar CT	Energy goes to the highest-load intervals, so the battery never runs empty before the peak
Automated pump shifting	Enrolled pumps skip their 18:00 run; LEO schedules it in the midday hour with most solar surplus (≈ 0.71 kWh per home per day)	Moves load out of the evening ramp every day, onto local solar
Automated AC events	On DFPO stress days enrolled ACs cycle to 40% for 2 h with pre-cooling; ≈ 11.3 kW per event	Large, reliable kW at the peak instant, worth paying ₹75 per home per event
SMS / IVR offers	Contextual bandit chooses ₹0, ₹25, ₹50 or ₹100 per household per event	Reaches homes without smart devices or smartphones; pays only where it buys a real cut
Pre-outage	On a DISCOM load-shedding notice, reserve what the backup premises need ($\times 1.5$) and keep shaving with the rest	Critical premises stay powered through scheduled cuts
Backup circuit	Inverters island onto their backup ports; premises served by priority with current limits; staggered re-transfer	Clinic, water and livelihoods keep running; no cold-load surge on restoration
Recommendations	Tap-change, inverter-setting and outage-scope notes to the DISCOM, with evidence	LEO advises; the DISCOM decides on its own network
Explainable console	Predicted events with reasons, every action with its why, outcome revealed against the forecast	Operators and DISCOM engineers can audit every decision
Transparent settlement	IPMVP baselines, random holdout, DR paid first, pooled rewards capped by realised revenue	Defensible to the independent verification agency; households never promised more than was earned

5.2 User journeys

Operator — the day before a hot April evening

At 19:30 the forecast predicts phase-Y undervoltage to 218 V and a transformer overload to about 206% of rating from 16:15. The console lists each predicted event with its causes and LEO's planned response: charge each block from midday solar, discharge 41 kWh per phase across 16:30–21:00, move 20 pumps to 10:45, call an AC event and SMS offers for 17:45–19:45. At 23:30 the operator approves. Next evening the console shows the plan executing, the live corrections, and the outcome against the forecast; residual undervoltage on long lines is escalated to the DISCOM as a tap-change recommendation.

Household with a pump or AC

The household enrolls once (app, SMS or IVR, with per-purpose consent^[34]), and a ₹2,090 smart plug or ₹999 IR blaster is fitted. The pump now runs at midday on neighbourhood solar; on about twenty summer evenings the AC is cycled for two hours after a pre-cool, with an

SMS beforehand and a one-tap override. It earns ₹100/month (pump) or ₹75 per event (AC), plus its share of the community pool, and its electricity bill is unchanged.

Critical premise during load shedding

BESCOM announces a 19:40–21:10 cut at 14:00. LEO enters Pre-outage, tops the blocks up to the backup reserve and SMSes all 150 homes. At 19:40 the inverters island and the clinic, borewell pump, tuition centre and shops stay powered from the backup circuit by priority; at 21:10 they are re-transferred to the grid in batches.

DISCOM engineer

The DISCOM dashboard shows an action queue — forecast undervoltage on phase B with a recommended tap raise, an outage with its scope — each with acknowledge/dispatch/resolve states and machine-readable JSON, plus verified kW for DFPO reporting.

5.3 The applications we built

LEO has three user-facing applications: one web app for the operator and the DISCOM, a citizen app, and SMS/IVR for households without smartphones. All are role-based (operator, DISCOM, household), read from the same backend, and run in the prototype today. The operator console was designed around one question: *what did LEO see coming, what did it do and why, and what actually happened?*

Table 3. User-facing applications built in the prototype.

Screen	Who	What it shows and does
Map control room	Operator	deck.gl map over real satellite imagery of the neighbourhood: feeder lines coloured by voltage, the transformer, battery blocks, sensors and backup premises. A timeline replays any recorded day, with a toggle between what happened and what was forecast the day before. Banners state the current situation in plain words.
What LEO predicted	Operator	Day-ahead forecast of transformer loading (forecast vs actual) and one card per predicted event: when it was predicted, why (load, temperature, irradiance), LEO's planned response, and the outcome once the timeline reaches it. Events the forecast missed are flagged.
What LEO did, and why	Operator	Time-ordered action log — forecast issued, plan approved, DR offers, pump shift, AC event, battery charge/discharge, mode changes, outages, escalations — each with its reason, filterable, click-to-peek on the timeline.
Scenario runs	Operator	Peak day with and without LEO, announced load shedding, unplanned outage, and a sunny surplus day with and without LEO; a glossary explains every term.
Plan review	Operator	Per-phase charge/discharge schedule and the DR offers for tomorrow; the operator approves before anything is dispatched.
Live operations	Operator	Battery state of charge and far-end voltage per phase.
DR events	Operator	Offers sent, accepted and held out, response rate and offers by amount (₹0/25/50/100).
Settlement and members	Operator	Daily ledger by stream with the payout scaling factor and per-household drill-down; outage-day backup fees; member registry (sanctioned load, business, critical premise).
Impact and economics	Operator, DISCOM	Section 11 and 13 results for every configuration: reliability against the baseline, monthly charts, who gains what, the deal zone, sensitivity and the assumptions register.
Action queue	DISCOM	Recommendations (tap-change raise, review inverter settings, outage scope) with time window, phase, evidence and machine-readable JSON; states open → acknowledged → dispatched → resolved.

Screen	Who	What it shows and does
Reporting	DISCOM	Verified flexibility against the random holdout, DFPO progress, registered critical premises, and a planning insight on recurring stress.
Home, Offer, Earnings	Household	Current earnings and the latest offer in plain language (“Earn ₹50 if you switch off your cooler or pump 7–9 pm tonight”); settled payments by stream (DR incentive, surplus absorption, discharge rebate, backup fee).
Usage	Household	15-minute import and export, explicitly labelled “last updated yesterday” because meter data arrives a day late.
Consent	Household	Per-purpose, revocable consent: meter-data access, DR offers, critical-premise disclosure, sensor hosting (DPDP Act).
Outage and Community	Household	Backup energy received during an outage (for registered premises); neighbourhood totals — total paid, battery status, outage minutes protected.

Inclusion. The app and SMS/IVR share one backend path for offers and responses. In the pilot, enrolment, consent, offers, earnings, load-shedding and restoration alerts, outage reporting and STOP are designed to work over DLT-registered SMS and an IVR in Kannada, Hindi and English, so households without smartphones are not excluded^[28, 35]. The prototype sends offers and alerts through a mock SMS service; the IVR is designed but not built.

Technology. Next.js 15 (App Router) with React 19 and Tailwind CSS 4; a deck.gl 9 map^[36]; a FastAPI service over PostgreSQL for all data; role-based access enforced at the route level; everything packaged with Docker Compose alongside the mock field devices. The prototype uses stub logins (operator, DISCOM, household); a pilot uses managed authentication with the same roles.

6. System architecture

The architecture follows one rule: the operator commands only assets it owns, and everything that must keep working during an internet outage runs on the neighbourhood edge gateway. Figure 1 shows the system context; Figure 2 the components by tier with the protocol on each link. Component numbers follow our System Architecture v3.0^[28], updated to the built system.

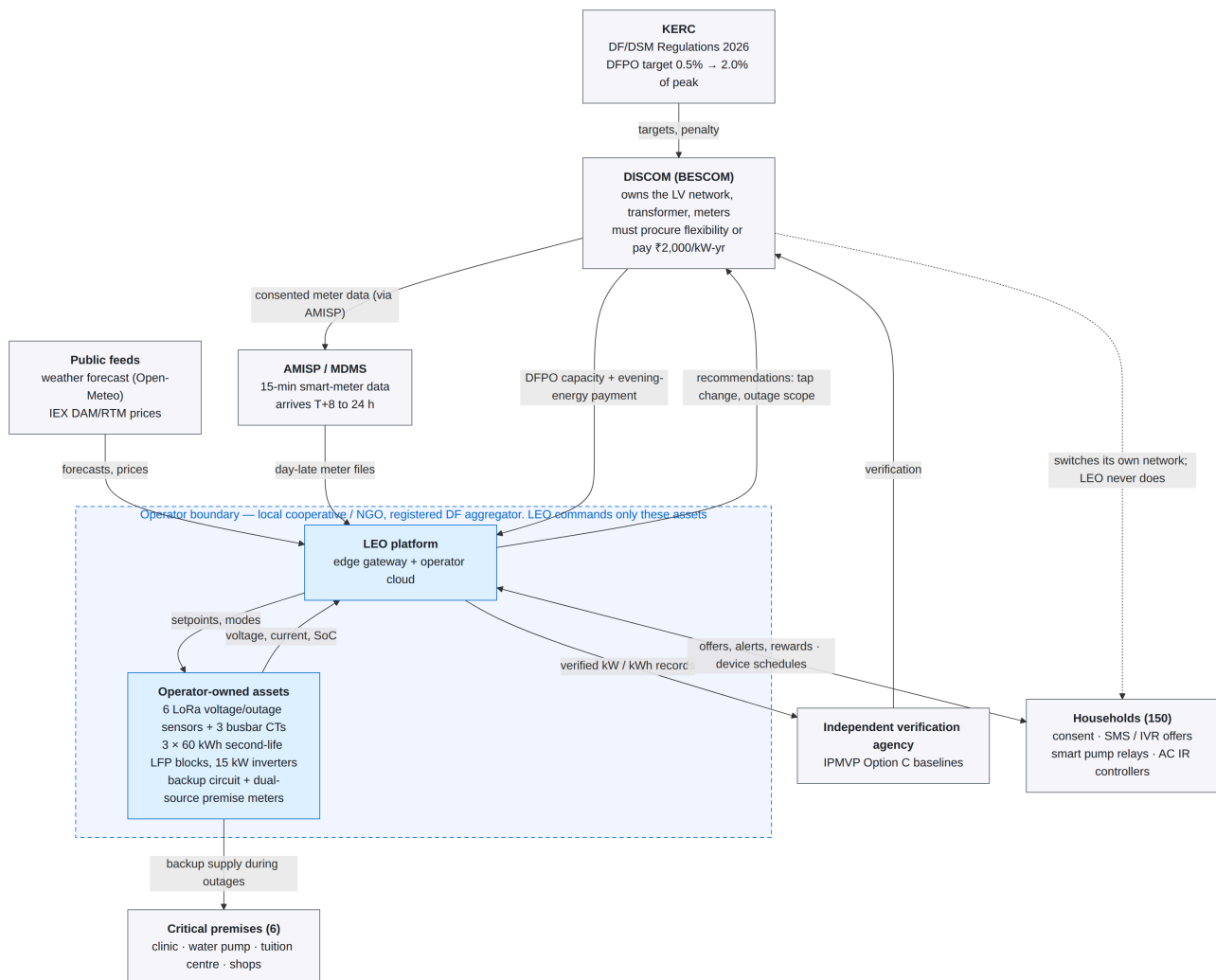


Figure 1. System context. The dashed boundary is the operator's: LEO commands only the sensors, batteries, backup circuit and its own messages. The DISCOM switches its own network; LEO only recommends. Money flows from the DISCOM to the operator under the DFPO contract and from the operator to households.

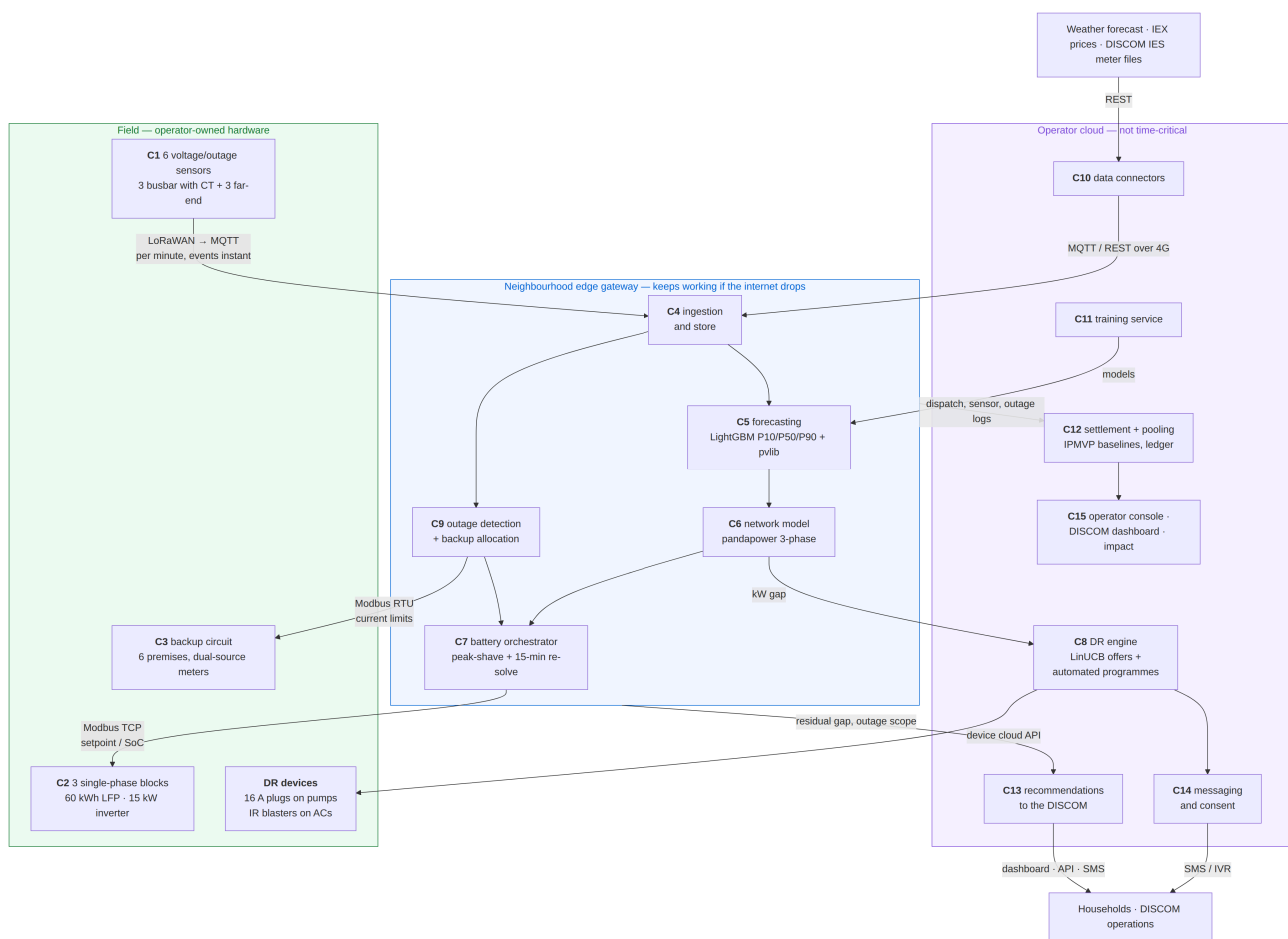


Figure 2. Layered architecture: field hardware, the neighbourhood edge gateway (offline-capable) and the operator cloud, with protocols on each link.

Table 4. Components.

#	Component	Tier	Role	Interface / protocol
C1	Voltage/outage sensors	Field	Per-phase RMS voltage every minute, power-fail/restore instantly; busbar units add a CT	LoRaWAN IN865 → ChirpStack → MQTT
C2	Three single-phase storage blocks	Field	Second-life LFP + BMS + 15 kW hybrid inverter (grid and backup ports)	Modbus TCP (SunSpec where supported)
C3	Backup circuit and premise meters	Field	Separate LV cable to 6 premises; dual-source meters with current limiters	Modbus RTU (RS-485)
—	Automated DR devices	Field (homes)	16 A smart plugs on pumps, Wi-Fi IR blasters on ACs	Wi-Fi via device cloud API
C4	Ingestion and store	Edge	Validates and timestamps every observation; single source of truth on the gateway	MQTT, Modbus polling
C5	Forecasting inference	Edge	LightGBM quantile load model + pvlib PV per phase	—
C6	Network model	Edge	pandapower 3-phase power flow: violations and safe charge/discharge limits	—
C7	Battery orchestrator	Edge	Day-ahead water-filling plan; 15-minute receding-horizon re-solve; Normal/Pre-outage modes	Modbus TCP to C2
C8	Demand-response engine	Cloud	Event days and windows; disjoint LinUCB offers; automated pump and AC programmes	SMS/IVR via C14; device API

#	Component	Tier	Role	Interface / protocol
C9	Outage detection and backup	Edge	Classifies scope from sensor pattern; Backup/Restoration modes; premise allocation	Modbus RTU to C3
C10	Data connectors	Cloud	Weather, IEX, SLDC, consented meter data	REST; India Energy Stack
C11	Training service	Cloud	Weekly load-model retrain, PV calibration	—
C12	Settlement and pooling	Cloud	IPMVP baselines, holdout verification, two-part revenue, DR payments, Streams 1-2	UPI payouts
C13	Recommendation engine	Cloud	Residual gaps and outages → structured DISCOM actions	Dashboard, API/webhook, SMS
C14	Messaging and consent	Cloud	DLT-registered SMS, IVR in Kannada/Hindi/English, per-purpose consent	SMS / IVR
C15	Consoles	Cloud	Operator console, DISCOM dashboard, impact and economics	HTTPS

6.1 Why three tiers

Field hardware is cheap, standard and operator-owned. **Edge** components are those whose failure would be unsafe or costly if the internet dropped — live battery control, outage detection and the backup circuit. **Cloud** components need history, people or money and tolerate hours of latency — training, DR offers, settlement, recommendations and the consoles. This split also makes multi-DT operation cheap: one cloud serves a cluster; each DT adds only a gateway and field hardware.

6.2 Energy and money flows

Energy flows are shown as a day in Figure 9 (Section 11) and money flows as a Sankey in Figure 10 (Section 13).

7. Technical approach

7.1 Design principle: plan day-ahead, correct live

Household meter data arrives 8–24 hours late^[19], DT meters rarely communicate^[17], and only the operator's own sensors are live. LEO therefore plans a day ahead from forecasts and corrects within the day using simple, transparent rules on its own measurements. Figure 3 puts the latency on every arrow.

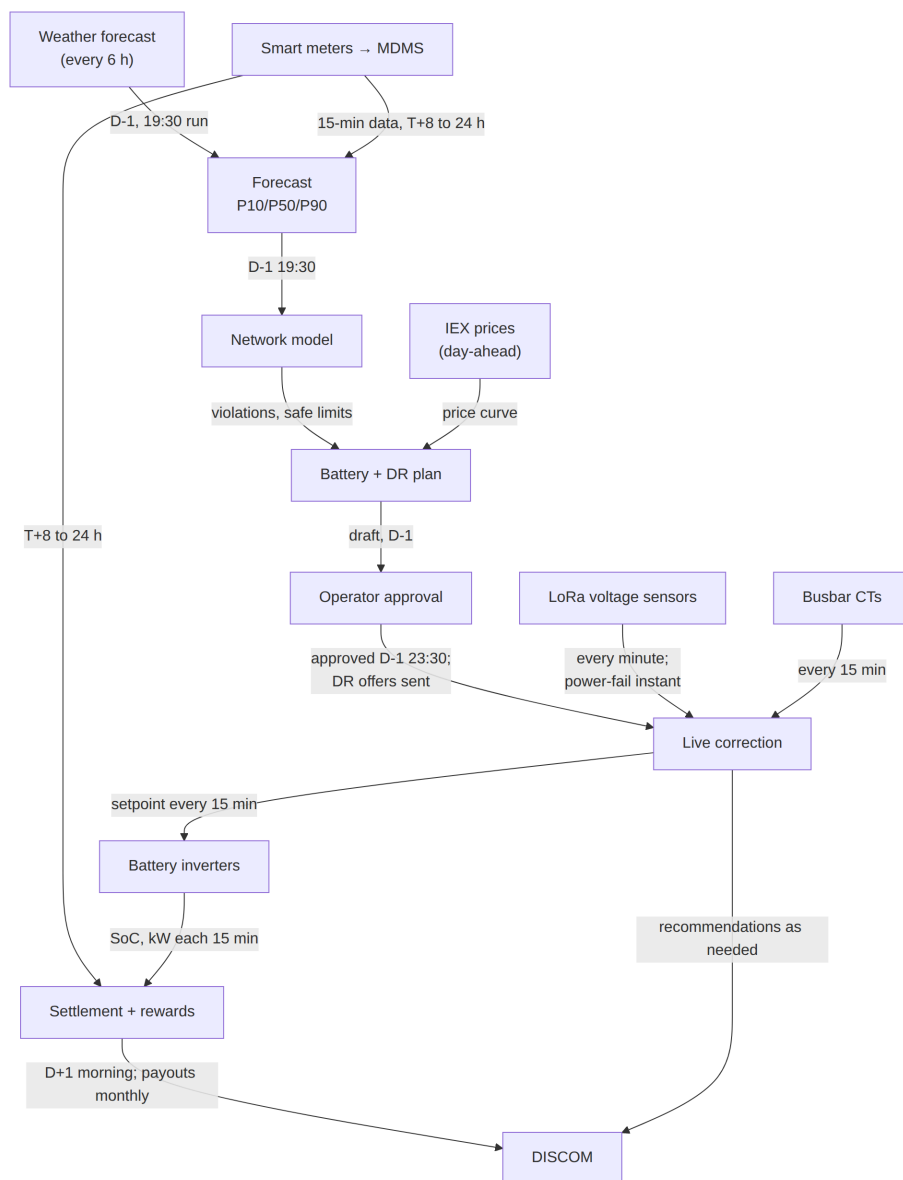


Figure 3. Data flow with latency on each arrow. Meter data is a day late, so LEO plans at D-1 19:30, the operator approves at 23:30, live correction runs every 15 minutes from LEO's own sensors, and settlement runs on D+1.

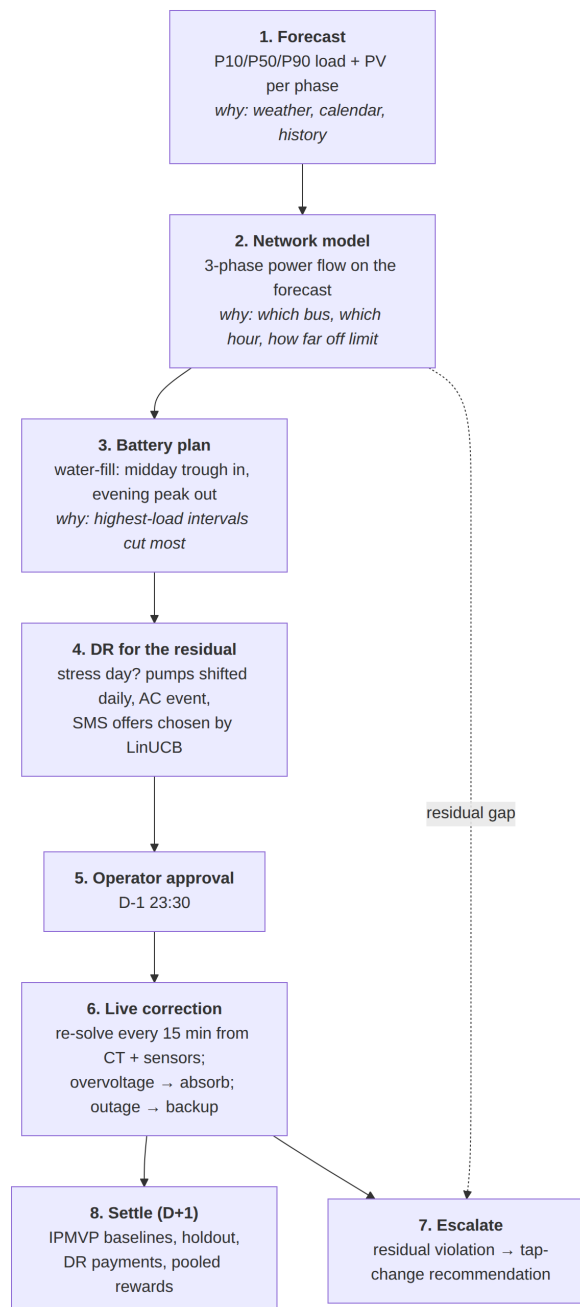


Figure 4. The decision pipeline. Each stage records its reason, which is what the operator console shows — the pipeline doubles as the explainability story.

7.2 Forecasting (C5)

A LightGBM quantile model^[37] predicts gross load per phase at P10/P50/P90 for 96 fifteen-minute intervals from weather forecasts (temperature, humidity, irradiance)^[38], calendar and holiday features, and lagged consumption that respects the meter-data latency rule — a feature is only used if it would have arrived by run time. Rooftop PV behind net meters is invisible, so a pvlib physics model^[39] is scaled by a calibration factor fitted on clear-sky middays and multiplied by each home's installed kWp; net load is gross minus PV. On weeks the model never saw in training (four quarterly folds), the evening feeder peak was forecast with a mean absolute percentage error of 1.9% and per-phase load with a 7.6% weighted error; actual load exceeded P90 5.1% of the time (nominal 10%), i.e. slightly conservative.

These are on synthetic data and will be wider in the field; Section 19 lists the validation we plan.

7.3 Network model (C6)

pandapower's unbalanced three-phase power flow^[40] runs on a feeder generated from the OpenStreetMap street graph^[41] (transformer, minimum-spanning-tree LV lines, households at real building footprints, round-robin phase assignment with deliberate imbalance). It returns per-bus, per-phase voltage and transformer and line loading, flags violations of the $\pm 6\%$ band, and gives each block's safe charge and discharge limits from the busbar voltage sensitivity, so a battery action never creates a violation of its own.

7.4 Battery orchestration (C7)

Both directions are water-filling problems on the phase's load curve: in the evening, find the level such that shaving everything above it uses exactly the energy the block holds; at midday, find the level such that filling everything below it uses exactly the energy the block can take — the lowest-load midday intervals are those where rooftop PV exports, so the block absorbs local surplus first. The same solver runs day-ahead on the P50 forecast (the plan the operator approves) and every 15 minutes on what is left of the window, with the phase load measured by the busbar CT and the forecast bias-corrected. A far-end overvoltage triggers extra charging. SoC stays within 15–90%. In Pre-outage the floor rises to $1.5\times$ the backup premises' allowance for the announced cut; in Backup the floor drops to 5%.

The reference-trajectory (IRT) planner, after Huang et al.

Huang et al. schedule storage with a *prediction-free, dual-reference online optimisation*: instead of trusting a multi-hour forecast, the battery tracks two learned reference signals — an **Informative Reference Trajectory** (a target state-of-charge curve built from similarity-weighted historical days) and a **real-time value benchmark**^[42]. We adopted this scheduling idea because LEO cannot rely on accurate long-horizon forecasts, and we did not adopt the paper's double-auction market (Section 9). LEO implements it in three parts:

1. **Trajectory library (offline)**. For every historical day and phase, a perfect-foresight linear program using that day's true load and PV, with charging priced at the ToD tariff and discharge valued at an IEX-proxy price weighted by the day's own net-load shape, gives the SoC trajectory the battery *would* have followed had it known the day in advance. In deployment the library is progressively replaced by realised operating history.
2. **Similarity blend (day-ahead)**. Tomorrow is matched to historical days using only what is known a day ahead — forecast mean and maximum temperature, a clear-sky index, day type and festival flag — with a Gaussian kernel on standardised features and a penalty for a day-type mismatch; the ten nearest days' trajectories are blended, weighted by similarity, into one reference SoC curve per phase. Household load is never used, because it is not available in time.
3. **Convex day-ahead plan**. A quadratic program (cvxpy/OSQP)^[43] minimises *ToD charging cost – flexibility value of discharge in forecast-violation intervals* + $\lambda \cdot \Sigma(\text{SoC} - \text{SoC}_{\text{IRT}})^2$, subject to SoC dynamics with round-trip efficiency, the battery's power rating, the network model's per-interval safe charge/discharge limits, the SoC window and the reserve floor in every interval. The deviation term keeps the plan close to what worked on similar days, so a forecast error moves the plan less.

Which planner ran in the evaluation. The year-long evaluation in Sections 10–11 used the water-filling planner: its single objective is exactly the one the DT needs (flatten the evening peak), it is closed-form and explainable, and the same solver runs the 15-minute live re-solve. The IRT planner is implemented and exercised on sample days; it becomes the default once ToD tariffs reach the operator's connection and realised history replaces the bootstrap library, because it adds price awareness and history-based robustness that water-filling lacks.

7.5 Demand response (C8)

Event days. DFPO is measured in kW at a single instance in the peak period^[7], so DR is reserved for the year's most stressed days — the top 5% by forecast evening peak, about 22 events a year, 2 hours each. Tata Power-DDL called 12–16 events a year, three to four a month in April–September^[44].

Automated programmes. 70% of pump owners (21 homes) enrol a smart plug: every day their pump moves from 18:00 to the midday hour with the most forecast solar surplus, with a 5% household override. 60% of AC owners (19 homes) enrol an IR blaster: on event days the AC cycles to 40% for two hours, pre-cools in the two hours before (30% of the curtailed energy) and rebounds slightly after (20%), with a 10% override. Each participating home is paid ₹75 per event; each pump home ₹100/month.

SMS / IVR offers. For homes without devices, a contextual bandit chooses a flat offer of ₹0 (appeal), ₹25, ₹50 or ₹100 per event, maximising expected value of the cut (kW at ₹2,000/kW-yr spread over the year's events plus the ₹7.92/kWh evening premium) minus the expected payout. We use disjoint LinUCB^[45] — one ridge regression per offer level — with context features including window load, appliance ownership, whether a shop is open during the window, and the household's separate response rates to paid offers and to appeals. It is pre-trained on a randomised pilot at real event times and offer cadence, then serves greedily with a small exploration bonus. Offers respect a 4-per-month cap and a 3-day gap; 10% of eligible homes are held out at random to verify the programme's causal effect. Reduced energy rebounds by 50% after the window.

Calibration against Indian pilots. Our simulated SMS acceptance is 23% against Tata Power-DDL's 56% average participation (with ₹250 payouts)^[44] — deliberately conservative. Our automated AC events deliver ≈ 0.08 kW per connected customer, inside the 0.03–0.16 kW range measured by BYPL and AEEE's automated DR pilot^[46].

7.6 Outages and backup (C9) and operating modes

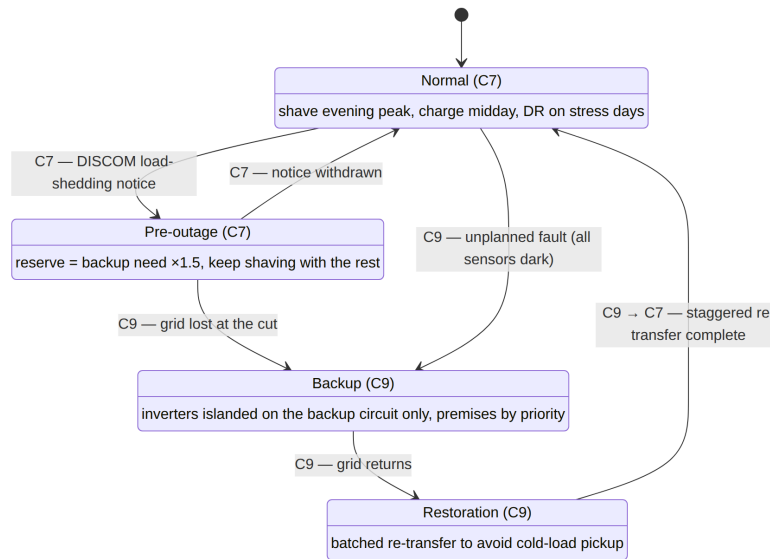


Figure 5. Operating modes, triggers and owning component. C7 owns Normal and Pre-outage because it holds the forecasts; C9 owns Backup and Restoration because it holds the events.

Power-fail messages from the six sensors are classified by pattern: all sensors dark → upstream fault; one phase → phase fault; one far-end only → local. Inverters anti-island automatically (IEC 62116 / IEEE 1547)^[47] — this is inverter behaviour, not an orchestrator decision — and then supply only their backup ports. C9 allocates each phase's available energy (SoC down to 5%, spread over the expected outage) to premises by priority (health 16 A, water 10 A, livelihood/education 6 A) and on restoration re-transfers them in batches to avoid a cold-load surge. Figures 6–8 show the three scenarios.

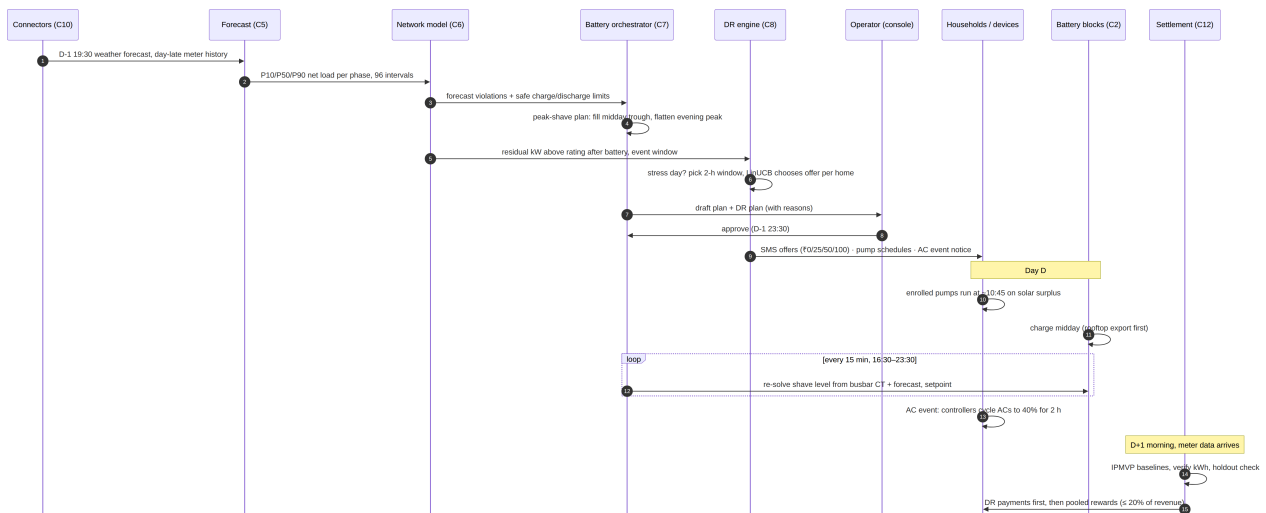


Figure 6. Sequence — peak day: forecast, plan, approval, offers and device schedules, live re-solve, settlement.

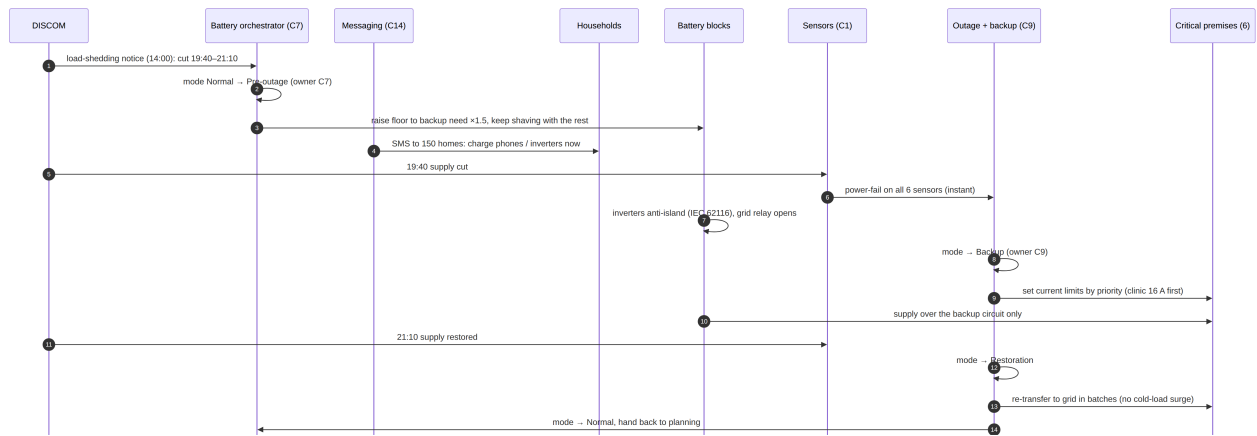


Figure 7. Sequence — announced load shedding: notice, Pre-outage, cut, anti-islanding, backup by priority, staggered restoration.

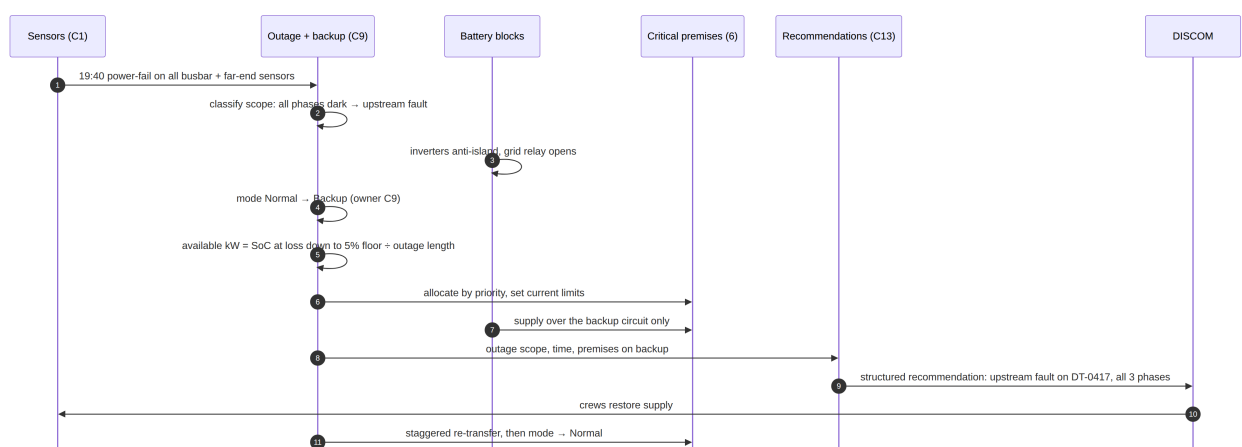


Figure 8. Sequence — unplanned upstream fault: detection and scope from sensors, backup from remaining SoC, recommendation to the DISCOM.

7.7 Settlement and pooling (C12)

Settlement runs once a day when meter data for the previous day arrives. Each household's reduction is its IPMVP Option C baseline (same window over similar non-event days, adjusted by that day's pre-event usage) minus actual consumption; the random holdout tests whether the programme caused the reduction and whether baselines are biased^[32, 33, 48]. Revenue is the two-part contract (Section 13) plus backup fees. Payouts are sized backwards from realised revenue: a share $\alpha = 20\%$ forms the payout budget; contractual DR payments are paid first; the rest funds Stream 1 (households exporting solar while the blocks charge, pro-rata) and Stream 2 (a rebate of the blocks' evening discharge, allocated down a participation-ranked queue with a per-household cap). If claims exceed the budget they are scaled pro-rata, never promised. No payment is ever negative. This follows sonnenCommunity's pooled model rather than a per-kWh auction^[27, 49] (Section 9).

7.8 Recommendations (C13) and explainability (C15)

When local resources cannot close a gap — typically undervoltage along long lines, which a battery at the busbar cannot fix — LEO sends the DISCOM a structured recommendation (e.g. raise the tap on phase B, 16:15–21:00, with forecast and residual evidence), and for outages, the detected scope. The operator console pairs each predicted event with its reasons, the actions LEO took and why, and the outcome compared with the forecast.

8. Standards, protocols and integration with existing infrastructure

LEO uses the standards Indian utilities and vendors already use, so moving from the prototype to a pilot changes addresses and drivers, not code. LEO talks to the simulator through MQTT, Modbus and REST exactly as it would to real devices^[28].

Table 5. Standards and protocols: current prototype use and the integration path.

Standard / protocol	Layer	Prototype use	Real-world integration
DLMS/COSEM (IS 15959, IEC 62056), IS 16444	Meter data model	Synthetic meter files keyed by OBIS codes, released on the SLA schedule by a mock endpoint	Household AMI data via the DISCOM's MDMS / India Energy Stack
India Energy Stack (Beckn, W3C DID, Verifiable Credentials)	Consented data exchange	Mock endpoint with the same schema	Consented meter-data exchange with DISCOMs (four pilot DISCOMs)
MQTT	Gateway message bus	Mosquitto broker for all telemetry	Same
LoRaWAN IN865-867, ChirpStack	Sensor radio	Simulated ChirpStack payloads	Sensors to the gateway concentrator; no SIM cost; works offline
Modbus TCP, SunSpec models	Inverter control	pymodbus mock server, three unit IDs	Three single-phase hybrid inverters; OpenEMS drivers where available
Modbus RTU (RS-485)	Premise meters	Mock devices	Dual-source meters with current limiters
IEC 62116; IEEE 1547-2018	Inverter grid behaviour	Anti-islanding modelled per block	Inverter certification; volt-var/volt-watt where supported
IEC 61968/61970 (CIM)	Network model exchange	Asset naming in recommendations	GIS / network-model exchange with the DISCOM
OpenADR 2.0/3.0; IEEE 2030.5	DR / DER signalling	Not used	DISCOM-to-aggregator event signals (both referenced by IES)
IEC 60870-5-104, DNP3, IEC 61850	DISCOM SCADA / substation	Not used	DISCOM-internal; LEO never writes to it
IPMVP Option C	Measurement and verification	Baselines and holdout in settlement	Verification by empanelled independent agencies (KERC)
TRAI TCCCPR 2018 (DLT)	Commercial SMS	Mock SMS service	Registered sender IDs and templates (₹5,900 one-time)
DPDP Act 2023	Personal data	Per-purpose consent model	Legal basis for data handling
UPI	Payments	Recorded in ledger	Monthly payouts

References: DLMS/COSEM and Indian meter standards^[50]; India Energy Stack^[51, 52]; MQTT^[53]; LoRaWAN and ChirpStack^[54]; SunSpec Modbus^[55] and OpenEMS^[56]; anti-islanding^[47]; CIM^[57]; OpenADR and IEEE 2030.5^[58]; IPMVP^[32]; TRAI DLT^[35, 59]; DPDP^[34]; CEA smart-metering interoperability^[60].

8.1 Integration with what the DISCOM already has

- **Meter data:** via the MDMS export or India Energy Stack, with consumer consent — LEO never needs live household data.
- **Network data:** GIS / consumer indexing gives the consumer→DT→feeder mapping and sanctioned loads; LEO's generated feeder is replaced by the DISCOM's model, exchanged in CIM terms.
- **Operations:** recommendations arrive on a dashboard, an API/webhook or SMS/email to the section office; the DISCOM acts in its own SCADA/OMS. DR events can later arrive from the DISCOM over OpenADR or IEEE 2030.5.
- **DFPO reporting:** verified kW per DT per event, with baselines and holdout, in a form an independent verification agency can reproduce.

9. Design decisions and alternatives we rejected

We grounded the design in one academic paper, two policy roadmaps and several industry deployments, and for each we recorded what we took and what we left out^[27]. The single filter every choice passed was: *minimal grid-level intervention, genuine affordability, useful to the DISCOM without new capex*.

9.1 What we adopted, and from where

- **Huang et al. (2025)** — scheduling storage against reference signals (a target SoC trajectory and a value benchmark) rather than relying on accurate long-horizon forecasts; *not* its double-auction market^[42].
- **Schneider EcoStruxure ADMS/DERMS** — the DERMS concept of aggregating many small distributed resources into one virtual resource, with attention to reverse power flow, hosting capacity and non-wire alternatives; *not* SCADA, OMS or closed-loop VVO/FLISR^[26].
- **sonnenCommunity** — pooled shared-capacity membership settled in an ordinary ledger; *not* Brooklyn Microgrid's blockchain auction^[49, 61].
- **CEEW / Centre for Net Zero roadmap** — DT meters are rarely communicating, so design around day-late AMI; a synthetic smart-meter repository is the recommended near-term strategy; consent architecture is a named gap; DFPOs and aggregators are the near-term market^[17].
- **RMI / distribution franchisee literature** — kept as a future ownership option only^[69, 71].

9.2 Decision register

Table 6. Key design decisions, alternatives and the evidence that forced them.

Decision	Chosen	Rejected	Why (evidence)
Settlement ledger	Plain relational ledger, Sonnen-style pooled model	Blockchain / DLT settlement	Brooklyn Microgrid (LO3 Energy) stalled at pilot scale; its retrospective and an empirical comparison both found a database faster, cheaper and better suited to small, periodic, locally-trusted settlements ^[61, 62] . UPPCL's P2P pilot was 12 participants with mock trading ^[63] .
Price mechanism	Pooled capacity with fair-access rules	Iterative P2P double auction between households	Auction convergence assumes price-elastic participants (e.g. microgrids with storage); a household's rooftop export is nearly fixed regardless of price ^[27, 42] .
DR learning	Contextual bandit (disjoint LinUCB)	Full reinforcement learning	The question — which offer to make this household now — has no long action sequences; fatigue and budget are captured as context features. Bandits converge with far less data and are standard for which-offer problems ^[27, 45] .
DR learner structure	One model per offer level	One shared linear model with level as a feature	The shared model paid shops that are open during the event and non-responders, and gave ₹0 to price-sensitive homes; separate per-arm models learn each offer's own response.
DR mechanism	Automated pump shifting and AC cycling + SMS offers	SMS behavioural DR only	A typical low-income home cuts ~0.17 kW by SMS; automated devices give reliable kW and daily shifting, and cost ₹999–2,090 ^[31, 46] .

Decision	Chosen	Rejected	Why (evidence)
DFPO revenue unit	kW at the peak instance + ₹/kWh evening energy	₹/kWh only	MERC measures flexibility in kW at a single peak instance and penalises shortfall per MW ^[7] ; most DISCOM value is avoided evening purchase, so an energy component lets the DISCOM share it.
Battery planner	Water-filling peak shave, re-solved every 15 minutes	Greedy rule: discharge flat out when voltage dips	The greedy rule emptied every block by 19:00 — before the worst interval. Water-filling spends energy on the highest-load intervals by construction.
Storage topology	Three single-phase blocks	One three-phase battery	Indian LV networks are unbalanced; a balanced three-phase inverter cannot relieve one phase. Three 10–60 kWh second-life packs are also easier to procure than one large pack.
Battery chemistry	Second-life LFP	New packs; NMC	Second-life packs cost 40–60% of new ^[64, 65] ; LFP has far better thermal stability in tropical heat, and fade flattens under gentle stationary cycling ^[66, 67, 68] .
Inverter size	15 kW per phase	7.5 kW per phase	Research prices one 7.5–15 kW unit at the same ~₹1.2 L; the larger rating cut the verified peak by ~10 kW more for no extra capex.
Sensors	Six LoRaWAN loggers + busbar CTs	Cellular (GSM/NB-IoT) per sensor	No SIM cost, no dependence on mobile operators, works when the internet is down; ESMI documented GSM disruptions ^[29] .
Live data	Operator's own sensors only	Live household smart-meter data	AMISP SLAs deliver 15-minute data 8–24 h late; designs that assume it live fail in the field ^[19, 21] .
Control authority	Advisory to the DISCOM; control only own assets	Full ADMS/SCADA, closed-loop VVO/FLISR	Requires field-device authority a third party should not claim; we keep the DERMS aggregation idea with a clean interface boundary ^[26] .
Ownership	Cooperative / NGO DF aggregator	Distribution franchisee	Bhiwandi cut AT&C from ~63% to 19–25% but several franchises were cancelled and the input-based model is criticised ^[69, 70, 71] ; the aggregator role exists in regulation today.
Simulation environment	Own closed-loop LV simulator (pandapower)	Grid2Op	Grid2Op targets transmission-level redispatch and French grids, not Indian LV demand-side flexibility ^[72] .

10. Evaluation method and baseline

No DISCOM data was available, so — as CEEW recommends while AMI matures^[17] — we built a synthetic but physically grounded world and ran LEO against it through the same interfaces it would use in the field.

10.1 The simulated neighbourhood

- **Network:** a real Hoskote (Bengaluru Rural) street grid from OpenStreetMap, one 100 kVA DT (433 V, 250 V phase-to-neutral, $\pm 6\%$ limits), 150 households at real building footprints across three phases^[41]. The DT has outgrown its rating, as many peri-urban DTs have.
- **Households:** bottom-up appliance loads every 15 minutes (base load, fridge, temperature-driven AC and cooler, pumps, shop hours) — median 210 kWh/month and 0.52 kW in the April evening; 25% with 2–6 kWp rooftop PV; 20% AC, 35% coolers, 15% pumps, 10% shops. Hidden behavioural personas drive DR responses and are never visible to LEO^[73].
- **Weather:** one real year of hourly Open-Meteo weather for Hoskote (Sept 2025 – Sept 2026); PV from pvlib with each home's hidden tilt, azimuth and soiling^[38, 39].
- **Outages:** about 50 per year — 70% upstream, 20% single-phase, 10% local — more frequent in the monsoon and the evening, lognormal durations with a 35-minute median, literature-typical because ESMI raw data was not downloadable^[29]; plus two announced 90-minute evening load-shedding cuts in each sampled summer week.
- **Latency:** meter data reaches LEO on the AMISP SLA schedule; LEO's live view is only its sensors, CTs and batteries.

10.2 Protocol

We replay one week in every month (days 8–14) — 84 days covering summer, monsoon and winter — under each configuration, scaling sums to a year. Every configuration sees the same world, weather, outage schedule and household draws, so differences come from LEO's decisions alone. The load forecast is retrained in four quarterly folds with the evaluated weeks held out, so no forecast is scored on data it was fitted to. Battery state of charge, transformer temperature and DR history carry over day to day within each week. Transformer ageing uses the IEEE C57.91 thermal model with first-order top-oil dynamics and the ageing acceleration factor^[74]; technical losses come from the power flow.

Table 7. Configurations evaluated against the baseline.

Configuration	What is enabled
Baseline	No battery, no DR, no backup circuit — same world, weather, outages and households
LEO software + SMS DR	Forecasting, network model, SMS offers; no storage, no devices
LEO software + automated DR	Adds smart pump plugs and AC IR blasters
LEO, 3 × 15 / 30 / 45 / 60 kWh	Battery blocks of each size (C/2 to C/4 inverters) + SMS DR + backup circuit
LEO, 3 × 30 / 60 kWh + automated DR	Battery + automated DR + SMS DR + backup circuit (recommended: 60 kWh)

10.3 Metrics

Table 8. Reliability metrics.

Metric	Definition
M1 Overload trips / outage hours	LV fuse trips from sustained conductor overload; customer-hours of outage, including expected DT-failure outages (failures/yr × 48 h repair × 150 homes)
M2 Voltage quality	Customer-hours outside $\pm 6\%$ (all buses) and far-end sensor minutes
M3 Critical-premise availability	Backup-served minutes ÷ outage minutes at registered premises
M4 Evening peak reduction	Transformer evening peak (16:30–23:30), average and worst day
M5 Verified flexibility	Mean cut in the DT's evening peak on the year's 10% most stressed days (the DFPO measurement days), kW
Transformer ageing	Equivalent ageing hours per year → expected life and failure rate

11. Measurable reliability improvement

All figures are per transformer per year, measured against the defined baseline — the identical world without LEO (Section 10). The recommended configuration is three 60 kWh blocks on 15 kW inverters plus automated DR.

443 h

premise-hours of outage kept powered (100% of critical-premise outage time vs 0%)

414

customer-hours/yr of DT-failure outages avoided (failure rate 5.8% → 0.1%)

44%

lower average evening peak (129 → 72 kW); worst day 167 → 108 kW

967 → 130 h

hours per year above rating; energy above rating 25,000 → 701 kVAh

-42%

customer-hours of undervoltage (59,085 → 34,227)

49,836 kWh

rooftop solar per year stored locally instead of exported (25% of export)

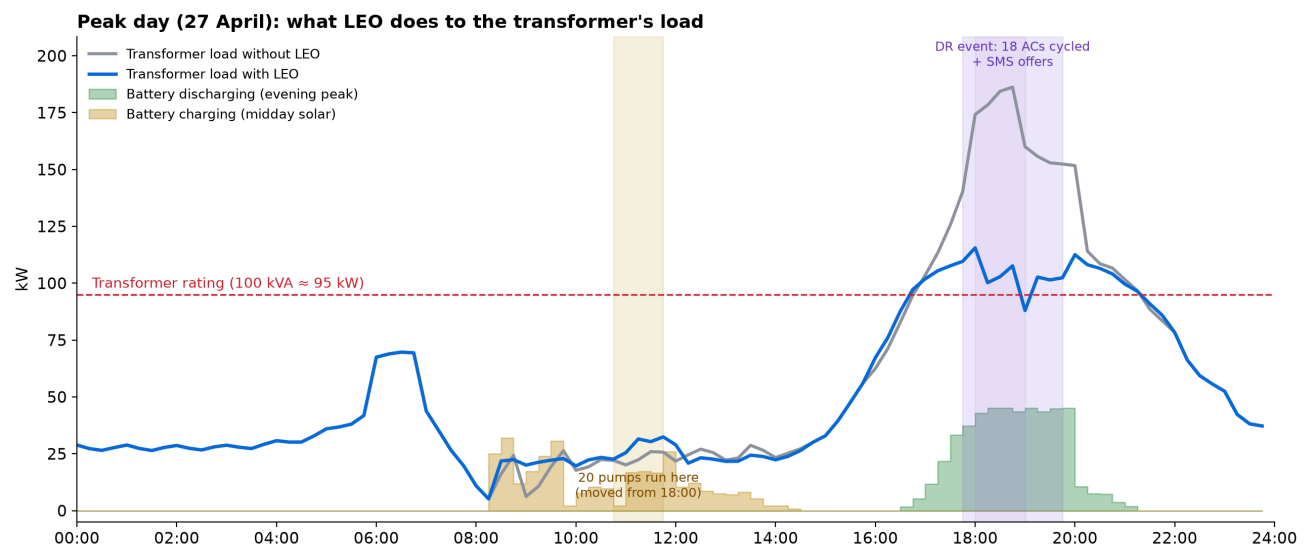


Figure 9. The intermittency bridge on the year's peak day (27 April): midday charging from rooftop solar, 20 pumps moved to 10:45, the evening discharge spread across the peak, and the automated AC event with SMS offers. The transformer peak falls from 186 kW to 116 kW.

Table 9. Reliability by configuration, per transformer per year (selected configurations).

Configuration	Evening peak avg (kW)	Worst day (kW)	Verified kW (M5)	Critical availability (M3)	DT failure rate	Hours overloaded	Undervoltage cust-h (M2)	Losses (kWh)
Baseline (no LEO)	129	167	0	0%	5.8%	967	59,085	11,890
LEO software + DR, no storage	129	166	1.4	0%	5.1%	967	59,137	11,882
LEO software + automated DR, no storage	116	145	19.7	0%	1.8%	951	47,023	11,290
LEO, 3 x 15 kWh blocks	108	150	20.5	87%	2.0%	937	54,219	11,550

Configuration	Evening peak avg (kW)	Worst day (kW)	Verified kW (M5)	Critical availability (M3)	DT failure rate	Hours overloaded	Undervoltage cust-h (M2)	Losses (kWh)
LEO, 3 x 60 kWh, 15 kW inverters	83	119	47.5	100%	0.2%	266	44,991	10,855
LEO, 3 x 60 kWh + automated DR	72	108	60.9	100%	0.1%	130	34,227	10,360

11.1 Reading the results

- **Outage hours.** LEO reduces outage hours through two channels: critical premises stay powered through faults and announced load shedding (443 premise-hours a year), and the transformer's expected failure rate falls from 5.8% to 0.1% a year, avoiding 414 customer-hours of multi-day replacement outages. M1 fuse trips did not occur in either arm on this feeder.
- **Supply during intermittency windows.** The evening ramp is exactly when solar disappears; LEO moves 49.8 MWh of midday solar and 5.5 MWh of pump load across it each year and keeps the average evening peak below the rating.
- **Where storage matters and where DR matters.** Automated DR alone cuts the evening peak by about 13 kW at very low capex; storage is what protects critical premises and the transformer; together they reach a 60.9 kW verified cut.

What LEO does not do, stated plainly. It cannot prevent upstream faults or load shedding, only prepare for them and protect registered premises; homes not on the backup circuit still lose supply. It cannot fix voltage drop along long LV lines from the transformer end — the remaining undervoltage is escalated to the DISCOM as tap-change recommendations. The results come from a synthetic world calibrated to public data and Indian pilots; field validation is the first step of the roadmap.

12. Ownership, operations and maintenance model

12.1 Who owns and operates LEO locally

A local cooperative, NGO or self-help-group federation registers with the DISCOM as a demand-flexibility aggregator — a role recognised by the MERC 2024 and KERC 2026 regulations^[6, 7, 23]. It owns the sensors, batteries, backup circuit, devices and the LEO software licence, and runs a cluster of about 20 DTs — for example one BESCOM O&M section or one gram panchayat. Households are members: they consent, respond to offers, and receive payments; they own nothing and risk nothing.

Table 10. Actors, ownership and control.

Actor	Owns	Controls	Does not control	Relationship
DISCOM	LV network, DT, meters (via AMISP), meter data	Switching, taps, crews, restoration	Operator assets	Buys verified flexibility under DFPO; receives recommendations; shares meter data with consent
Operator (cooperative / NGO aggregator)	Sensors, batteries, backup circuit, devices, LEO	Its batteries, backup circuit, devices, messages, ledger	DISCOM supply; household bills	Runs LEO; earns DFPO payments; pays households
Households (150 per DT)	Appliances; consent	Whether to respond or override	Anything else	Members; receive DR payments and pooled rewards
Critical premises (6)	Their loads	—	—	Pay a backup subscription and per-kWh fee
Independent verification agency	—	Verification	—	Reproduces LEO's baselines from DISCOM meter data

12.2 Operations

- **Staff:** one skilled technician per cluster (Karnataka minimum wage for a skilled electrician, Zone 1: ₹32,145/month^[75]), recruited locally — visual inspections, inverter filter cleaning, sensor and device swaps, enrolment drives, community relations.
- **Daily:** the operator reviews and approves tomorrow's plan in the console (about 10 minutes per DT), answers recommendations status from the DISCOM, and handles household queries by phone/IVR.
- **Platform:** cloud hosting ≈₹5,000/month per cluster; DLT-registered SMS at ≈₹0.145 per message plus ₹5,900 one-time registration^[14, 59, 76].

12.3 Maintenance

Table 11. Maintenance plan.

Asset	Routine	Life / replacement	On failure
Second-life LFP packs + BMS	Monthly BMS health and temperature review; temperature-aware depth of discharge in summer	min(7 years, 2,500 equivalent full cycles) — about 7 years at 336 cycles/yr; replacement budgeted	BMS limits or refuses commands; block isolated, others continue

Asset	Routine	Life / replacement	On failure
Hybrid inverters	Quarterly filter cleaning, firmware updates	10+ years	Inverter reverts to safe default after a command timeout
LoRa sensors and CTs	Quarterly check; battery-backed for last-gasp	5 years	Gateway flags missing data; forecast-only operation
Smart plugs / IR blasters	Self-reporting; household-swappable	5 years	Household reverts to manual; excluded from events until fixed
Backup circuit and premise meters	Annual inspection with the DISCOM	20+ years (cable)	Premise falls back to grid only
Edge gateway	Remote monitoring; UPS	5 years	Cloud flags gateway offline; blocks run their last plan, then safe default

12.4 Legal and regulatory basis

- **Demand flexibility:** KERC DF/DSM Regulations 2026 — targets, aggregator role, verification by empanelled agencies^[6, 24].
- **Backup circuit:** supplying other premises needs a licence unless exempt; Section 13 of the Electricity Act allows exemption for cooperatives and NGOs, and Section 14 allows supply without a licence in notified rural areas — hence the preference for notified rural or peri-urban sites^[77].
- **Grid connection:** each block needs DISCOM connection approval and certified anti-islanding^[47].
- **Data and messaging:** DPDP Act consent; TRAI DLT registration^[34, 35].

12.5 Governance

The household share α of revenue, the backup fee and the DR offer levels are set by the cooperative's members each year, within the bounds the unit economics allow (Section 13). The ledger is open to members; payouts can never exceed realised revenue and are never negative.

13. Unit economics and affordability

Unit economics are computed by multiplying the simulated physical quantities (kW verified, kWh shifted, hourly grid import, transformer ageing, backup minutes, DR offers and payouts) by prices in one assumptions file, each tagged with its source in the model's assumptions file. Recommended configuration: three 60 kWh blocks on 15 kW inverters plus automated DR, per transformer, per year, in a cluster of 20 DTs.

13.1 Capital cost

Table 12. Capital cost. Battery at ₹5,000/kWh (second-life, 40-60% of new^[64, 65]); inverters ₹1.2 L per 7.5-15 kW unit; sensors + concentrator ₹30,000; gateway ₹25,000; backup meters + cabling ₹4,000 per premise^[14]; IR blaster ₹999, smart plug bundle ₹2,090^[31].

Item (per DT)	₹
Sensors and gateways	₹59,500
Smart relays controllers	₹62,871
Battery packs	₹9.00 L
Inverters	₹3.60 L
Backup circuit	₹24,000
Installation	₹1.69 L
Registration setup	₹10,295
Total per DT	₹15.85 L

13.2 Operating cost

Table 13. Operating cost. Technician and cloud shared across the cluster; M&V agency cost sits in the DISCOM's DSM portfolio.

Item (per DT per year)	₹
Technician	₹19,287
Cloud connectivity	₹4,800
Sms	₹613
Insurance	₹15,854
Maintenance	₹23,781
Mv agency	₹0
Total per DT per year	₹64,336

13.3 Operator profit and loss

Table 14. Operator P&L. Contract: ₹2,000/kW-yr for 60.9 verified kW plus ₹5.00 per verified evening kWh (50,630 kWh/yr); energy settlement on the operator's ToD connection^[78, 79]; backup fees.

Line	₹ per year
Dfpo capacity	₹1.22 L
Evening energy	₹2.53 L
Energy settlement	₹87,264
Backup fees	₹16,733

Line	₹ per year
Revenue	₹4.79 L
Paid to households (DR first, then pooled share)	−₹95,786
Operating costs	−₹64,336
Net cash per year	₹3.19 L
Payback on capex	5.0 years
10-year NPV at 10% (battery replacement and salvage included)	₹1.10 L

13.4 DISCOM

Table 15. DISCOM ledger vs no LEO. Power purchase is priced hour by hour — ₹10/kWh in 18:00–23:00, ₹1.91 at midday, ₹7.6 otherwise^[2, 3] — so round-trip losses, DR rebound and network losses are netted out. Failures valued at ₹5.05 L per 100 kVA DT^[14].

Line	₹ per year
Power purchase saved	₹3.52 L
Dfpo penalty avoided	₹1.22 L
Dt failures avoided	₹29,003
Paid to operator capacity	−₹1.22 L
Paid to operator energy	−₹2.53 L
Energy settlement with operator	−₹87,264
Retail revenue lost to dr	−₹2,108
Gross saving	₹5.03 L
Net after paying the operator	₹38,890

The DISCOM's **gross** saving is ₹5.03 L per DT per year — ₹50.32 Cr per year across 1,000 overloaded DTs. It reduces technical losses on this DT by 1,529 kWh/yr (12.9%) and stops buying 50.6 MWh of evening power a year. Most of the gross saving flows to the operator under the contract because the operator carries the capital, devices and household payments; the split is a negotiated point inside the deal zone below.

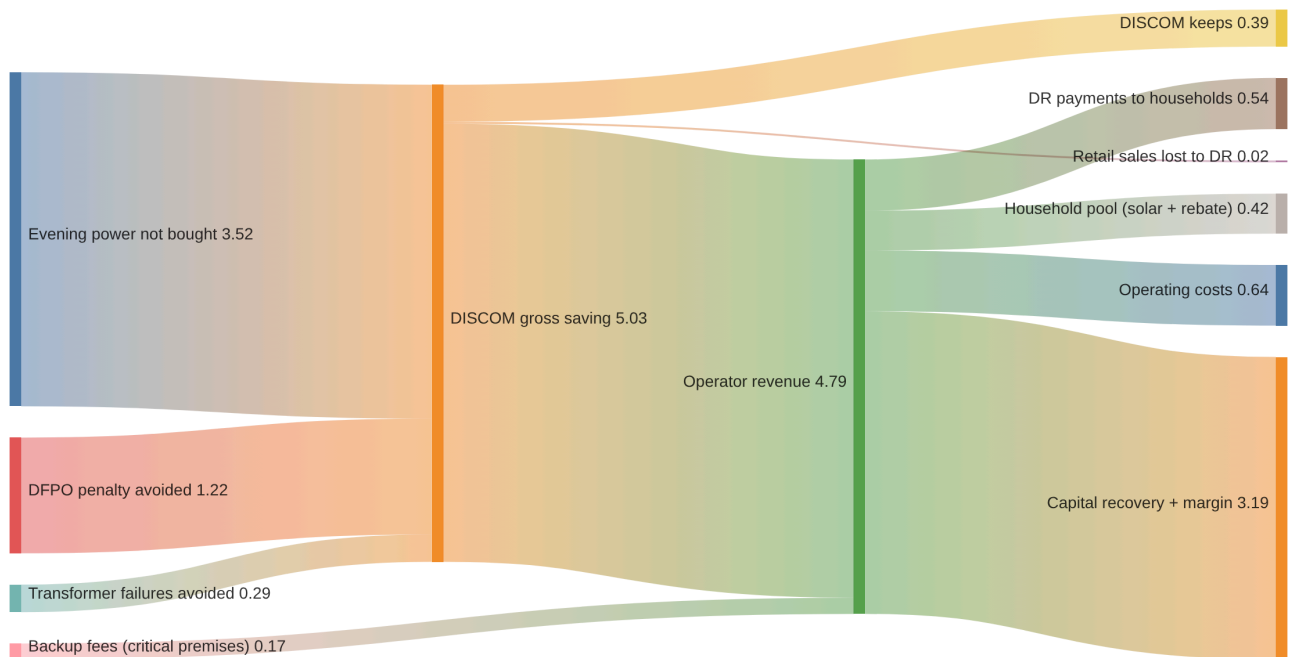


Figure 10. Money flow per transformer per year (₹ lakh): the DISCOM's gross saving, the contract payment to the operator and what the DISCOM keeps; the operator's revenue split into DR payments, the household pool, operating costs and capital recovery.

13.5 Is there a deal?

Payments between the operator and the DISCOM cancel out, so a deal exists exactly when their combined value is positive. For the recommended design the operator breaks even at ₹4.56 per verified evening kWh and the DISCOM gains up to ₹5.77; the contract rate of ₹5.00 sits inside, and the combined 10-year value is ₹3.49 L. Smaller batteries and SMS-only DR do not close (Figure 11).

Is there a deal? Operator break-even (amber) vs the most the DISCOM can pay (blue)

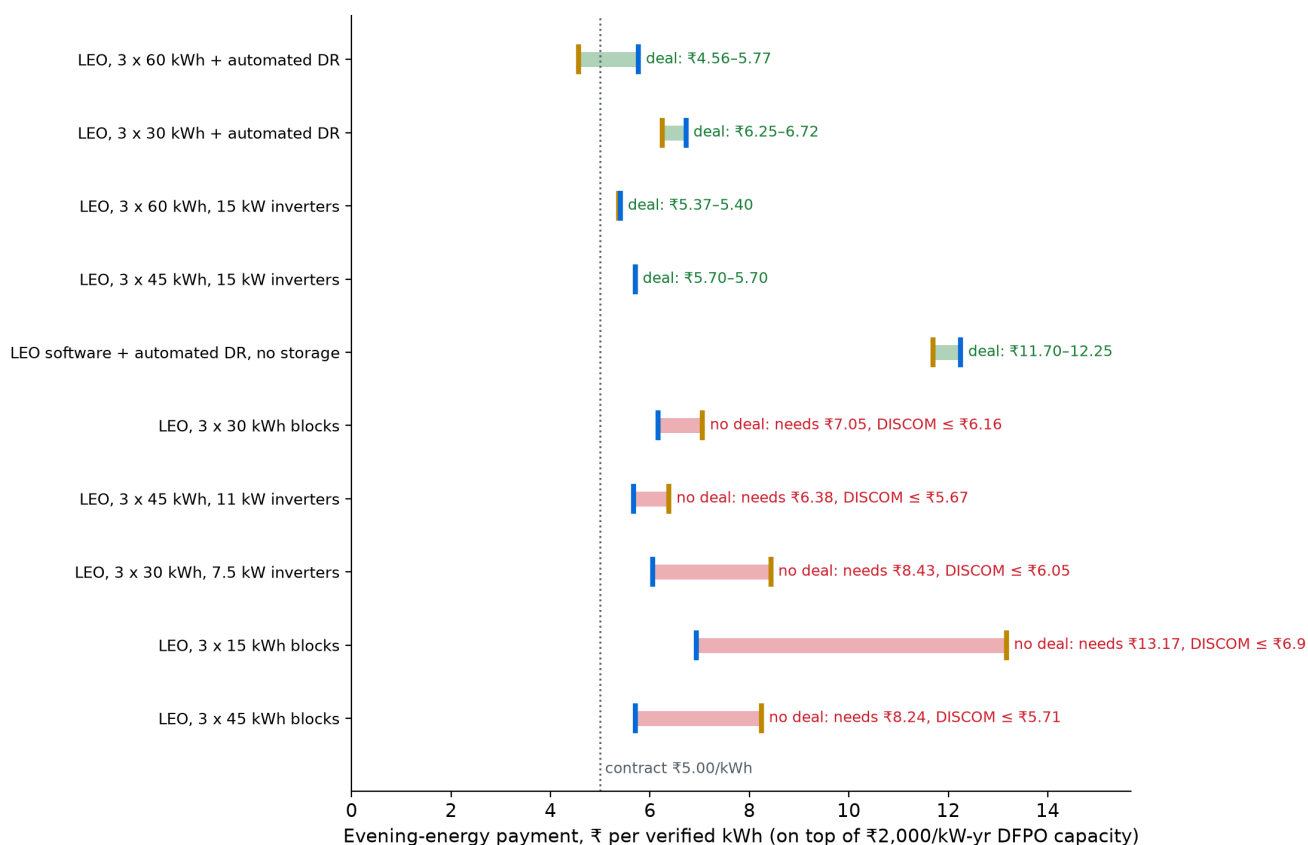


Figure 11. Deal zone by configuration: the operator's break-even evening-energy rate (amber) against the most the DISCOM can pay (blue), on top of the ₹2,000/kW-yr DFPO capacity payment.

13.6 What has to stay true

Table 16. Thresholds, one at a time, at which the arrangement stops being worth doing.

Uncertain input	Combined value stays positive if	Assumed
DFPO shortfall penalty	$\geq ₹1,067/\text{kW-yr}$	₹2,000 (MERC)
DISCOM evening power price	$\geq ₹8.73/\text{kWh}$	₹10.00 (IEX cap)
Second-life battery price	$\leq ₹6,223/\text{kWh}$	₹5,000
Insurance + maintenance	$\leq 6.1\%$ of capex	2.5%

Table 17. Sensitivity of the recommended design, one assumption at a time.

Change	Operator 10-yr NPV	DISCOM net / yr	Payback (yr)
Energy rate -30%	−₹2.63 L	₹1.15 L	6.1
Energy rate +30%	₹4.83 L	−₹37,056	4.2
Battery price -30%	₹5.38 L	₹38,890	3.9
Battery price +30%	−₹3.18 L	₹38,890	6.1
Cycle life -30%	−₹54,286	₹38,890	5.0
Cycle life +30%	₹1.10 L	₹38,890	5.0
Peak power price -30%	₹1.10 L	−₹95,498	5.0
DFPO penalty -50%	₹1.10 L	−₹22,001	5.0
Cluster of 5 DTs, not 20	−₹3.37 L	₹38,890	6.4
30% capital grant	₹5.86 L	₹38,890	3.5

Change	Operator 10-yr NPV	DISCOM net / yr	Payback (yr)
Household share 0.5	−₹7.73 L	₹38,890	9.1

13.7 Affordability for low-income households

Table 18. Household benefit. Upfront cost ₹0; electricity bills unchanged; no penalties.

Who	Per year	How
Pump home (21 homes)	₹2,027	₹100/month fee + ₹546 ToD bill saving (if ToD reaches LT domestic) + pool share
AC home (19 homes)	₹1,774	₹75 per event $\approx$ 20 events + pool share
Every other home	₹282	Pool share (solar absorption and discharge rebates) + any SMS offers accepted
Critical premise	backup at ₹12/kWh + ₹150/month	vs ₹15,000+ per kWh for a lead-acid home inverter replaced every 3–4 years, or a diesel genset

For comparison, Tata Power-DDL's pilot paid ₹250 per event (with ₹50/₹100 tiers) and ran 12–16 events a year^[44] — about ₹2,250 a year for a household attending nine events; LEO's participating homes earn in the same range. Households are paid from the operator's realised revenue at a share $\alpha = 20\%$; a deal remains possible up to $\alpha \approx 29\%$. A low-income household that could not afford a ₹15,000 inverter gets reliability on its street and income, with nothing to buy^[14].

14. Expected impact

Table 19. Impact, scaled linearly to 1,000 DTs similar to the simulated one.

Per DT per year	Per 1,000 overloaded DTs per year
₹5.03 L DISCOM gross saving	₹50.32 Cr
50.6 MWh of evening power not bought at ₹10/kWh	50.6 GWh
60.9 kW verified flexibility	60.9 MW towards DFPO
1,529 kWh technical losses avoided	1.53 GWh
DT failure rate 5.8% → 0.1%	≈57 fewer DT failures (₹2.90 Cr)
414 customer-hours of failure outages avoided	414 thousand customer-hours
443 premise-hours of critical supply kept	443 thousand premise-hours
49.8 MWh rooftop solar used locally	49.8 GWh
≈₹2,027 per participating household	150 thousand households reached

Context. Overload drives about 29% of BESCOM's ~38,000 annual DT failures^[4, 5]; Karnataka's DFPO target is 0.5% of peak in FY 2026-27, rising to 2.0% by FY 2029-30^[6]. Scaling assumes DTs as overloaded as the simulated one; on healthier DTs the transformer-failure value is smaller, which is why we target overloaded peri-urban DTs first (Section 18).

Social. Clinics, water supply, tuition centres and small shops keep running through cuts; households earn without buying anything; the community's own rooftop solar powers its evening.

15. Innovation

- **Built around day-late data.** Most designs assume live household data. LEO is engineered for the AMISP reality: plan a day ahead from forecasts, correct live only from six cheap sensors and the batteries' own CTs^[19].
- **Per-phase everything.** Three single-phase second-life blocks, per-phase forecasts, per-phase safe limits from a three-phase power flow — matched to unbalanced Indian LV networks.
- **One solver for plan and live.** The same water-filling problem produces the approved day-ahead plan and the 15-minute correction, so the operator approves what actually runs and the battery cannot run empty before the peak.
- **DR designed for the DFPO.** Flexibility is bought where the regulation values it — kW at the peak instant — with automated devices for reliable kW, daily pump shifting for energy, and a bandit that pays real money only where it buys a real cut.
- **A deal, not a subsidy.** A two-part contract (DFPO capacity + evening energy) derived from the DISCOM's own avoided costs, with a deal zone the model computes and the operator and DISCOM can negotiate within.
- **Honest settlement.** Payouts sized backwards from realised revenue, random holdouts for causal verification, no blockchain and no penalties.
- **Explainability as a feature.** Every predicted event has reasons; every action has a why; outcomes are compared with forecasts — auditable by the operator, the DISCOM and the verification agency.
- **Evidence-first.** Every price is from a tariff order, market data, a regulator, a DISCOM report or an Indian pilot, or is flagged for verification; the simulation is calibrated against Tata Power-DDL and BYPL pilot results^[44, 46].

16. Prototype: what we built and what we simplified

The full system runs locally with one command (`docker compose up`): mock field devices, the edge gateway, the operator cloud and the web consoles, against a closed-loop simulator. Recorded runs replay the year's peak day, an announced load-shedding day, an unplanned outage and a sunny low-demand day, each with and without LEO.

16.1 Technology stack

Table 20. Prototype technology stack.

Layer	Prototype	Why
Simulation	Python: numpy, pandas, pandapower (3-phase power flow), pvlib, osmnx, Open-Meteo history	Open, Python-native, no commercial solver ^[38, 39, 40, 41]
Forecasting	LightGBM quantile models, pvlib with clear-sky calibration	Fast, accurate on tabular data, quantiles for risk
Planning	Water-filling peak-shave planner; cvxpy/OSQP reference planner	Closed form, explainable; convex upgrade path
DR	Disjoint LinUCB; persona simulator	Learns with little data; personas never visible to LEO
Field interfaces	Mosquitto MQTT; pymodbus TCP/RTU mock battery and meters; mock SMS; mock IES	Same protocols as the field
Store and APIs	PostgreSQL; FastAPI	Relational ledger; simple REST
Consoles	Next.js; deck.gl map over satellite imagery ^[36]	One web app for operator and DISCOM
Evaluation	Year-sampled sweep across configurations; economics from one assumptions file	Reproducible; prices change without re-simulation

16.2 Prototype versus real deployment

Table 21. What we simplified for the hackathon and what replaces it in the field.

Area	Prototype	Real deployment
Household meter data	Synthetic, released on the AMISP SLA schedule by a mock IES endpoint	IES exchange or DISCOM export with consent
Voltage sensors	Simulated from power-flow voltages with noise and dropouts	LoRa loggers (ESMI-style), hosted with consent
Storage	Battery models behind a pymodbus server	Second-life packs, BMS and hybrid inverters over Modbus TCP
Backup circuit	Simulated premise meters	Cable, changeover, dual-source meters; legal basis secured
Network topology	Generated from OSM street geometry	DISCOM GIS / consumer indexing
Household behaviour	Hidden personas calibrated to pilots	Real households
Smart devices	Modelled pump shift and AC cycling	Commercial smart plugs and IR blasters via their APIs
SMS / IVR	Mock service	DLT-registered provider; IVR in Kannada, Hindi, English

Area	Prototype	Real deployment
Payouts	Ledger only	UPI
DFPO revenue	Contract rates in the assumptions file	Contract under the DISCOM's DF programme
Gateway	Docker Compose on a laptop	Raspberry Pi-class device on UPS at the battery site

16.3 Simplifications to be aware of

- The neighbourhood is synthetic (real streets and weather, generated households); outage rates are literature-typical, not fitted to ESMI.
- Load forecasts are scored on a synthetic world, so field accuracy will be lower.
- DR responses come from calibrated personas; AC curtailment is modelled from appliance duty cycles.
- Annual figures scale one week per month; load growth and battery fade over the horizon are modelled only through replacement.

17. Data and model design

All state lives in one PostgreSQL schema shared by the gateway and the cloud; every table carries a run identifier so recorded runs, evaluations and the live system never mix. The ER overview (Figure 12) is generated directly from the schema definition.

Table 22. Data model by group (PostgreSQL; contracts/ddl.sql).

Group	Tables	Key fields	Written by	Read by
Network and registry	neighbourhood, bus, line, household, appliance, sensor, battery_block, premise_backup, consent	bus_id, phase, sanctioned load, PV kWp, critical class, per-purpose consent	world builder / DISCOM GIS	network model, DR engine, consoles
Runs	run	run_id, LEO on/off, seed, period	simulator / gateway	every table (foreign key)
Observations	sensor_reading, meter_interval, battery_telemetry, premise_meter, event	ts_end, voltage, supply present, import/export kWh (day-late), SoC, backup kWh, event kind	sensors (MQTT), MDMS/IES, inverters (Modbus)	gateway, settlement, consoles
Forecast and network	forecast, network_result, phase_limit	P10/P50/P90 per phase, per-bus voltage and loading, is_forecast flag, safe kW	forecasting, network model	planner, explainability, recommendations
Decisions	plan, dispatch, mode_transition	planned and actual kW, SoC, rule triggered, mode and owner, approval	battery orchestrator, outage module	settlement, consoles
Demand response	dr_event, dr_offer	window, target kW, offer (₹), predicted/verified kWh, holdout flag	DR engine, settlement	bandit training, settlement
Settlement	ledger, period_revenue	entry type, amount (paise, never negative), scaling factor, revenue lines, α	settlement	households, DISCOM, verifier
Recommendations	recommendation	issue, action, phase, evidence, status (open → resolved)	recommendation engine	DISCOM dashboard
Evaluation	eval_sweep, eval_day, eval_economics	configuration, day, metrics, priced results	evaluation pipeline	impact page, this document

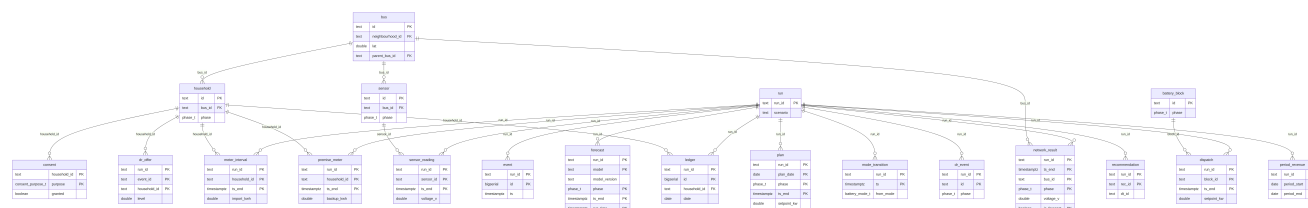


Figure 12. Entity-relationship overview of the core tables, generated from contracts/ddl.sql (the full 29-table diagram, 10_data_model_full.svg, is in the repository).

Table 23. Models.

Model	Inputs	Output	Validation
Load forecaster (LightGBM quantile)	Weather forecast, calendar, holidays, lagged load (latency-respecting), phase static features	P10/P50/P90 gross load per phase, 96 intervals	Held-out quarterly folds: evening peak MAPE 1.9%, phase WAPE 7.6%, P90 exceedance 5.1%
PV model (pvlib + calibration)	Irradiance, temperature, installed kWp	PV per home and phase	Clear-sky midday fit on net-import data
Network model (pandapower 3-phase)	Bus loads, battery setpoints	Voltages, loadings, violations, safe limits	Power-flow convergence every interval
Battery planner (water-filling)	Forecast or measured phase load, SoC	Setpoints per 15 minutes	Energy conservation; never below reserve
DR bandit (disjoint LinUCB)	Household, engagement, event context	Offer level and expected kWh per home	Holdout comparison; offer-response cross-check against pilots
Transformer ageing (IEEE C57.91)	Phase loading, ambient temperature	Hot-spot temperature, ageing hours	Standard thermal model

18. Implementation roadmap

Table 24. Implementation roadmap.

Phase	Timing	Scope	Exit criteria
0 — Prototype	Done (hackathon)	Full system on simulated world; year-sampled evaluation; unit economics	This document
1 — Shadow pilot	Months 0–6	One overloaded peri-urban DT in a notified rural/peri-urban BESCOM area; 6 sensors + CTs; consented meter data via MDMS/IES; LEO forecasts and recommends only	Forecast accuracy on real data; sensor uptime; DISCOM acceptance of recommendations
2 — Active pilot	Months 6–15	Install three second-life blocks, backup circuit (6 premises), smart plugs/IR blasters in enrolled homes; DR events in the April–September season; IPMVP verification with an empanelled agency	Verified kW on DFPO days; critical-premise availability; payouts settled; safety approvals
3 — Cluster	Months 15–30	20 DTs in one O&M section under one cooperative operator; two-part DFPO contract with BESCOM; one technician	Operator cash-positive; DT failures tracked against the section's history
4 — Scale	Year 3+	Multiple clusters and DISCOMs; OpenADR/IEEE 2030.5 event signals from the DISCOM; convex planner; additional devices (geysers, EV chargers)	DFPO portfolio share; replication playbook

Site selection. Start where the value is highest and the law clearest: DTs with a history of overload failures, in notified rural or peri-urban areas where Section 14 allows the backup circuit, with high rooftop-solar or pump penetration^[77].

Financing. Capital can be raised against the DFPO contract; capital grants (state or central storage support, CSR, or DISCOM co-funding from loss-reduction programmes) shorten payback further — a 30% grant raises the operator's 10-year NPV to ₹5.86 L.

19. Future scope

What we propose happens in the real world next:

- Replace the synthetic world with real DT data: GIS topology, MDMS/IES meter history, measured sensor voltages; retrain and re-validate forecasts.
- Field-validate DR: enrolment rates, override rates and kW per home for smart plugs and IR blasters, against the Tata Power-DDL and BYPL results^[44, 46].
- Negotiate the two-part contract with BESCO inside the computed deal zone; verify DFPO measurement rules under KERC.
- Confirm the legal basis for the backup circuit (Section 13 exemption or Section 14 rural proviso; whether storage counts as generation)^[77].
- Upgrade planning to the convex reference-trajectory planner with ToD costs and degradation^[42, 43]; temperature-aware depth of discharge for second-life packs^[68].
- Midday DR offers to absorb solar surplus (water heating, pumping) where overvoltage is the binding problem; EV charging via OCPP.
- Receive DR events from the DISCOM over OpenADR / IEEE 2030.5; publish verified kW to the DISCOM's DSM cell automatically.
- Multi-operator, cross-neighbourhood settlement — the point at which distributed-ledger guarantees might start to earn their cost^[27].
- Federated learning for forecasting across DTs without moving household data.

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